

The role of ICT in early childhood education: Scale development and research on ICT use and influencing factors

Stephanie Kerckaert^{a,b*}, Ruben Vanderlinde^{a,c} and Johan van Braak^a

^aDepartment Of Educational Studies, Ghent University, Ghent, Belgium; ^bHogeschool – Universiteit Brussel, Brussels, Belgium; ^cResearch Foundation Flanders, Brussels, Belgium

Nowadays, more and more authors are convinced that ICT (information and communication technology) in early childhood education provides multiple possibilities for young children. The current study, conducted in Flanders, aims to get a clearer picture of ICT use in Flemish preschools. For this purpose, a questionnaire was composed consisting of newly developed and existing scales. Based on the answers of 232 preschool teachers, two types of ICT use are distinguished in early childhood education, 'ICT use supporting basic ICT skills and attitudes', and 'ICT use supporting contents and individual learning needs'. 'ICT use supporting basic ICT skills and attitudes' occurs more frequently and is related to the grade of the preschoolers, teachers' self-perceived ICT competences and the number of years of experience with ICT at school. 'ICT use supporting contents and individual learning needs' is strongly related to the grade of the preschoolers, teachers' self-perceived ICT competences, ICT professional development and teachers' attitudes towards the possibilities of ICT for teachers in early childhood education. This indicates that professional development is a crucial factor in stimulating ICT use that transcends teaching basic ICT skills and attitudes.

Keywords: ICT; technology use; early childhood education; scale development; professional development

Introduction

Debate on ICT use in early childhood education

ICT (information and communication technology) use in early childhood education used to be a controversial topic. Some authors viewed ICT as a threat to playful learning and children's development (Cordes and Miller 2000; Healy 2003). They argued that ICT use leads to lack of exercise, isolated lives, poor concentration, impaired language development, etc. Currently, other authors promote the use of technology and argue that technology is a useful tool for learning (Bolstad 2004; Hatzigianni and Margetts 2012).

They point to a growing body of evidence which underlines the positive effects of ICT in early childhood. According to Hatzigianni and Margetts (2012) ICT 'presents a new space for exploration and discovery to young children, offers challenging activities and responds to children's curiosity' (5). Bolstad (2004) indicates that ICT already has an effect on the people and environments surrounding young children's learning and, as such, that these technologies offer new opportunities to strengthen many aspects of early childhood education practice (e.g. it can stimulate creativity and play, cognitive

*Corresponding author. Email: steph_kerckaert@hotmail.com

development, social interaction, etc.). Notwithstanding this debate, research on ICT use in early childhood education is still in its infancy.

Current context

The current study is conducted in Flanders, the northern region of Belgium. In 2007, an ICT curriculum for compulsory education was launched by the Flemish government (see Vanderlinde, van Braak, and Hermans 2009 for more information). This compulsory ICT curriculum is written in terms of ICT attainment targets and only applies to primary and secondary education (ranging from the age of 6 to 18, Vandenbroucke 2007). This means that preschools (ranging from age 2.5 to age six in Flanders and consisting of three grades) are not obliged to use or integrate ICT in their curriculum. The Ministry of Education considers it the responsibility of the preschools themselves to choose the moment to start working with ICT with their children (Vandenbroucke 2007). However, this does not mean that preschoolers do not have experiences with ICT. Teuwens (2011) indicates that preschoolers often have their first encounters with the Internet at home. In the Netherlands, McKenney and Voogt (2010) conclude that playing games is the computer activity young children most frequently do, both at home and at school. In a study in the UK (Marsh et al. 2005) parents report that 53% of children aged zero to six use a computer at home on a daily basis. As McKenney and Voogt (2010) state: ‘There is little dispute that today, children are using ICT even before they know how to read and write’ (656). Surprisingly, there is currently not much information available on the actual use of ICT in early childhood education. Existing publications typically concentrate on the possibilities and dangers of ICT for preschoolers (Bolstad 2004; Clements and Sarama 2003; Cordes and Miller 2000; Plowman and Stephen 2003; Siraj-Blatchford and Siraj-Blatchford 2005; Van Scoter, Ellis, and Railsback 2001; Yelland 2005), and rarely present empirical research data. What teachers think about the use and the place of ICT in preschools, the extent to which they already use ICT, and the factors which are related to this use, are rarely examined. Hence, the overall goal of this explorative study is to get a clearer picture of ICT use in early childhood education.

Possibilities of ICT in early childhood education

Plowman and Stephen (2005, 2006) indicate that there are ‘cultural’ differences between learning in the preschool sector and learning in primary schools. The curriculum is less prescriptive and there are different norms of professional practice. There is more emphasis on learning through play and less reference to formal and adult-directed teaching. In this way, integrating ICT into the early childhood curriculum can mean something completely different from integrating ICT into primary or secondary education (Campbell and Scotellaro 2009). Whereas in formal education, it is assumed that ICT can make education more child-centred, early childhood education is already to a large extent determined by this. In Plowman and Stephen’s study (2005), this child-centeredness in early childhood education translates into the use of the computer as one of many possible activities in free play. During free play, preschoolers can choose when or if they want to play with the computer.

While teachers give support in other activities and organise guided activities, this does not happen in activities with technology, as teachers seem to be afraid to give too much instruction, and so disadvantage the child-centred pedagogy (Plowman and

Stephen 2005). In a study of Morgan (2010), there is indeed evidence that the use of an interactive whiteboard does not necessarily lead to playful or interactive learning experiences. The teacher is in control and the technology is mostly used for instruction. Both studies indicate that, when using ICT, teachers need to be aware of their role in guiding the children. When ICT is used by the teacher, it risks supporting a more instructionist form of pedagogy, while harming the child-centred pedagogy (Morgan 2010). When ICT is merely used as a free activity, children tend to get frustrated and quickly proceed to another activity (Plowman and Stephen 2006). As Terreni (2010) states: 'Free play does not guarantee effective or creative engagement or development and there is still need to support and guide children's interactions in informed ways' (93).

Given the specific character of preschools, outlined above, the question as to what ways ICT can be used in early childhood education is important. ICT can be defined as 'anything which allows us to get information, to communicate with each other, or to have an effect on the environment using electronic or digital equipment' (Siraj-Blatchford and Siraj-Blatchford 2003, 4). ICT includes much more than just computers, it also includes digital cameras, creativity and communication software and tools, the Internet, telephones, fax machines, interactive stories, computer games, programmable toys, video conferencing, electronic whiteboards, etc. (Bolstad 2004).

Different authors give an overview of the possibilities of ICT for young children (e.g. Bolstad 2004; Clements and Sarama 2003; Kalas 2010; Siraj-Blatchford and Siraj-Blatchford 2005). These possibilities can be listed in five broad categories. Firstly, ICT can add an extra dimension to the play activities of young children (Bolstad 2004). Children can use ICT in realistic and imaginative socio-dramatic role-play, whilst learning the correct vocabulary and learning to use different forms of ICT. Secondly, ICT can contribute to both the language development and mathematical thinking of young children, through easy transitions between words and pictures and practice software, drawing programmes or computer manipulatives (Bolstad 2004; Kalas 2010). Further, ICT can provide unique opportunities for scaffolding and supporting children with special learning needs, or children from culturally or linguistically diverse backgrounds (Bolstad 2004; Kalas 2010). Next, when ICT is used in spontaneous learning and playing in the classroom, it can be a catalyst for social interaction (Clements and Sarama 2003), although adult guidance is often needed to gain the most from ICT environments (Kalas 2010; Siraj-Blatchford and Siraj-Blatchford 2005). Finally, Wood et al. (2008) acknowledge the motivational aspects of ICT. The speed, colours, dynamic presentation and instant feedback attract children. This illustrates that ICT offers multiple possibilities and can be embedded in early childhood education in many divergent ways.

ICT use and influencing factors

This study not only aims to get a clearer picture of ICT use in early childhood education. Since ICT integration in education is a slow and complex process that is influenced by many factors (Inan and Lowther 2010), it is useful to explore which factors are important in understanding ICT use in preschools. Because there is not much research available in the particular context of early childhood education, the following list is mainly based on research in primary and secondary education and focuses on teacher characteristics that may influence the use of ICT in the classroom. A distinction is made in different categories:

Background variables

Teachers' grade. The grade in which a teacher teaches is possibly an important factor in explaining ICT use in early childhood education. Teuwens (2011) concludes that the age of a preschooler is related to whether a child has experience with the Internet at home. Other authors stress the importance of the concept of 'developmental appropriateness' (Bolstad 2004; Kalas 2010; Siraj-Blatchford and Siraj-Blatchford 2000). This concept refers to the idea that the use of ICT should be appropriate within the scope of a child's development. This means that experiences with technology should be challenging, but also attainable for most children of a certain age (Siraj-Blatchford and Siraj-Blatchford 2000). Teachers are responsible for examining which forms of use are appropriate for a certain age and a certain child.

Teachers' experience in education. Age is frequently mentioned as a factor related to ICT integration (Hermans et al. 2008). Inan and Lowther (2010) include both age and years of teaching in their study and conclude that these demographics negatively affect the integration of ICT by teachers.

General teacher characteristics

Innovativeness. Van Braak (2001) describes innovativeness as a relatively-stable, socially constructed and innovation-dependent characteristic that indicates an individual's willingness to change his or her familiar practices. Innovativeness seems to exert an influence on ICT use in the classroom (van Braak 2001; van Braak, Tondeur, and Valcke 2004; Tondeur, Valcke, and van Braak 2008). These findings underpin the role of innovativeness as an important determinant in explaining the use of computers in classrooms (Tondeur, Valcke, and van Braak 2008).

Teacher self-efficacy. Tschannen-Moran and Woolfolk Hoy (2001) define teacher self-efficacy as 'a teacher's judgment of his or her capabilities to bring about desired outcomes of student engagement and learning, even among those students who may be difficult or unmotivated' (783). According to Sang et al. (2010) teacher self-efficacy is related to the use of ICT, the attitudes towards ICT and the teachers' ICT competences. Mueller et al. (2008) found different results. They conclude that all participants estimate their teacher self-efficacy highly, whether they use ICT in the classroom or not.

ICT-related teacher characteristics

Teachers' ICT competences. Self-perceived ICT competences reflect an individual's beliefs about his or her capabilities to use ICT (Compeau, Higgins, and Huff 1999). Hew and Brush (2007) indicate that a lack of knowledge and skills has been identified as a major barrier to ICT integration. They distinguish three types of knowledge and skills necessary to integrate ICT in the classroom: specific technology knowledge and skills, technology-supported-pedagogical knowledge and skills, and technology-related-classroom management knowledge and skills. Compeau, Higgins, and Huff (1999) and Sang et al. (2010) conclude that the degree to which teachers find themselves competent in integrating ICT in their class is an important factor related to the use of ICT in primary education.

ICT professional development. Vanderlinde and van Braak (2010) describe ICT professional development as an important strategy that fosters ICT integration. According to Galanouli, Murphy, and Gardner (2004), various approaches have been used,

including the dissemination of good practice, the investment in infrastructure and equipment and training programmes designed to raise skill levels and foster positive attitudes among teachers towards ICT. In a study of van Braak, Tondeur, and Valcke (2004), the amount of training undertaken by teachers was significantly related to the use of the computer in the classroom.

Experience with ICT at home and at school. Both van Braak (2001) and Hermans et al. (2008) conclude that experience with ICT at home and at school is a significant factor in explaining the classroom use of ICT. Van Braak (2001) conducted his study in secondary education, while Hermans et al. (2008) used teachers in primary education. Tsitouridou and Vryzas (2004) found similar results in early childhood education.

Attitudes towards ICT in the classroom. Several authors (e.g. van Braak, Tondeur, and Valcke 2004; Inan and Lowther 2010; Mueller et al. 2008; Sang et al. 2010) include attitudes towards ICT in their attempt to explain the use of ICT in the classroom and find a significant relation. According to Inan and Lowther (2010) teachers' attitudes towards ICT consists of teachers' perceptions of technology's influence on student learning and achievement on the one hand. On the other hand, it includes teachers' perceptions of the impact of technology on classroom instruction and learning activities.

Research purpose

The purpose of this study is twofold. Given the specific character of early childhood education, no appropriate measurement scales are available in the literature to study ICT use in early childhood education. Therefore, the first aim of this study is to develop a valid and reliable instrument that measures preschool teachers' ICT use. The second aim of this study is to explore which factors are related to ICT use in early childhood education. Background variables, general teacher characteristics and ICT-related teacher characteristics are used to explain ICT use in early childhood education. Where needed, new instruments are developed for the independent variables.

Research design

Participants

An online questionnaire was sent to 58 schools in Flanders, the northern region of Belgium. The constructed sample was representative for the Flemish preschool population based on province and school net. In total, 232 respondents filled in the questionnaire. This involves almost exclusively female teachers (99.6%), representing the actual distribution in Flemish male and female preschool teachers. Of the participants, 93.1% are teachers in early childhood education. 42.1% of the respondents teach in a first grade, 25.3% in a second grade, and 29.2% in a third grade. The other participants are presumably teachers who don't have their own class or teach in a class for children aged 2.5 to three years. Of these, 1.7% are care coordinators, 0.9% are GOK-teachers (GOK: Gelijke Onderwijskansen (Dutch); Equal educational opportunities, English), 0.4% are ambulant teachers and 3.9% perform other functions at their school, such as gym teacher. The participants have an average of 19.9 years of experience in education, ranging from 1 to 41 years ($SD = 9.82$).

Instruments

The questionnaire developed for this study consisted of a combination of existing and newly developed scales in order to formulate an answer to the research questions.

Dependent variable

As stated above, there are cultural differences between preschools and primary schools (Plowman and Stephen 2005, 2006). This specific character of early childhood education ensures that ICT can play a specific role within this context. Given the lack of appropriate scales in the literature related to ICT use in this particular context, a new scale was constructed for the dependent variable in this study, labelled as 'ICT use in early childhood education' (see 'Results' section). The new items were composed based on consultations with early childhood education teachers ($n = 4$) and specialists (like educational advisors and teacher trainers, working in the context of early childhood education, $n = 4$). These experts were asked to provide ways in which ICT can be used in early childhood education. Based on their comments an item pool was developed in order to construct a scale to measure ICT use in early childhood education.

Independent variables

Existing scales were used for the variables teachers' ICT competences, ICT professional development, innovativeness and teacher self-efficacy. The number of years of experience with ICT at home and at school, and the number of years of teaching experience were also recorded (see Table 1).

Besides the new scale for ICT use, a new item pool was developed to construct a scale to measure teachers' attitudes towards the possibilities of ICT in early childhood education, since there were no appropriate scales available in the literature. A distinction was made between attitudes towards the possibilities for teachers and attitudes towards the possibilities for preschoolers. More information on the factor analyses resulting in these new variables can be found in Appendix 1: Construction of the

Table 1. Existing scales.

Measurement scale	Description	Authors
Teachers' ICT competences	Measures the degree to which teachers find themselves competent in integrating ICT into their classroom practice	Vanderlinde and van Braak (2010)
ICT professional development	Assesses the extent to which teachers keep up with developments in the field of ICT integration, like taking part in in-service training programmes	Vanderlinde and van Braak (2010)
General innovativeness	Measures the willingness of teachers to introduce new methods and ideas into their teaching	Van Braak (2001)
Teacher self-efficacy	Measures the degree to which teachers believe in their capabilities to bring about desired outcomes of student engagement and learning	Tschannen-Moran and Woolfolk Hoy (2001)

independent variables ‘Attitudes towards the possibilities of ICT in early childhood education’.

Data analysis

SPSS 19.0 was used for the data analysis which was structured around three steps. First, exploratory factor analysis was conducted to identify the number of factors in preschool teachers’ ICT use and attitudes in early childhood education. Next, summary statistics were calculated for the independent and dependent variables, and Cronbach’s alpha was calculated to assess their reliability. Scales are reliable if the Cronbach’s alpha is greater than .70 (Mortelmans and Dehertogh 2008). Third, the influence of general and ICT-related teacher characteristics on ICT use was investigated by conducting a hierarchical regression analysis. This analysis makes it possible to examine the additional contribution of logically connected subsets of variables on the dependent variable.

Results

Construction of the dependent variable

A new scale was developed to explore ICT use in early childhood education. The original scale consisted of 20 items. The participants were asked to judge how often they use a certain kind of ICT on a Likert-scale with values 0 = never; 1 = every trimester; 2 = monthly; 3 = weekly; 4 = daily. Firstly, an analysis on item level was executed. Two items were excluded from further analysis due to high skewness and kurtosis. Next, an exploratory factor analysis (maximum likelihood with varimax rotation) was performed to determine the dimensionality of the scale. Six items were removed due to loadings across factors, one item was deleted due to low communality value. A solution consisting of two factors was obtained (Table 2). The first factor can be summarised as ‘ICT use supporting basic ICT skills and attitudes’ (n items = 5). This construct measures the ICT use that aims at teaching preschoolers how to handle ICT, and at developing a

Table 2. Factor analysis for ‘ICT use in early childhood education’.

	Factor	
	1	2
In my class I teach my pupils ICT basic skills.	.92	
In my class preschoolers use ICT to play games.	.89	
I teach preschoolers to use the computer and corresponding peripherals.	.89	
I teach preschoolers a positive attitude towards ICT.	.85	
I teach my pupils to handle ICT in a safe and responsible way.	.82	
I use ICT with preschoolers to illustrate certain subjects.		.85
I use ICT during instruction.		.73
My pupils discover ICT because I offer them digital stories.		.67
In my class I use educational websites to deliver certain contents to my pupils.		.66
I use ICT for remediating tasks for the preschoolers.		.58
In my class preschoolers with learning difficulties use adapted software/websites.		.46
R ² (proportion of variance explained)	38.5%	28.5%
α	.97	.86

positive attitude towards ICT. The second factor measures the concept '*ICT use supporting contents and individual learning needs*' (n items = 6). Within this construct ICT is not an end in itself, but a means to transmit other contents during instruction or free play and to anticipate the needs of preschoolers. This means that ICT can be used in individualised ways to support children with specific learning difficulties. These factors together explain 67% of the total variance. The Cronbach's alphas indicate that both scales are reliable (Table 2). Based on these results, the dependent variable needs to be handled as a two-factor construct.

Psychometric properties of the independent and dependent variables

Summary statistics for the independent and dependent variables are presented in Table 3. To facilitate interpretation of the results, all items were summarised into sum scores ranging from a minimum score of 0 to a maximum score of 100. Furthermore, Cronbach's alphas were calculated for the independent variables. All measurement scales show good internal consistency scores (from .80 to .97). Besides these scales, the number of years experience in education and with ICT at home and at school are used in the analyses. The participants had an average 19.9 years of experience in education ($SD = 9.82$) and indicated that they had 11.54 years of experience with ICT at home ($SD = 4.82$). At school, preschool teachers had used ICT for approximately 6.44 years in their classrooms ($SD = 4.32$).

Table 4 presents a correlation matrix. This offers a preliminary view of the relationships between the variables. In this study, the focus lies on the correlations with the two forms of ICT use. Table 3 shows a certain coherence between ICT use and ICT-related teacher characteristics and attitudes. Further, there are also relationships between the general and ICT-related teacher characteristics.

Table 3. Summary statistics of the independent and dependent variables.

Scale	Number of items	α	Mean	SD
Background variables				
<i>Grade</i>	-	-	-	-
<i>Experience in education</i>	-	-	19.9	9.82
General teacher characteristics				
<i>Teacher self-efficacy</i>	12	.90	80.09	7.14
<i>Innovativeness</i>	5	.88	72.71	16.49
ICT-related teacher characteristics				
<i>Teachers' ICT competences</i>	5	.84	58.09	19.83
<i>ICT professional development</i>	4	.86	41.52	20.88
<i>Experience with ICT at home</i>	-	-	11.54	4.82
<i>Experience with ICT at school</i>	-	-	6.44	4.32
Attitudes towards ICT				
<i>Possibilities of ICT for teachers in early childhood education</i>	9	.88	57.46	18.26
<i>Possibilities of ICT for preschoolers</i>				
- Basic ICT skills and attitudes	3	.80	77.34	14.16
- Development and learning process	6	.86	66.49	14.98
ICT use in early childhood education				
<i>Supporting basic ICT skills and attitudes</i>	5	.97	56.09	37.12
<i>Supporting contents and individual learning needs</i>	6	.86	29.49	24.45

Influencing factors

Given the results of the factor analyses on ICT use in early childhood education, two hierarchical regression models are presented: with '*ICT use supporting basic ICT skills and attitudes*' on the one hand, and '*ICT use supporting contents and individual learning needs*' on the other hand. These hierarchical regression analyses are conducted with only those respondents who have a full classroom responsibility and teach in first, second or third grade ($N = 209$).

ICT use supporting basic ICT skills and attitudes

Table 5 presents the results of a hierarchical regression analysis conducted to investigate which factors influence ICT use of teachers in early childhood education to support the basic ICT skills and attitudes of their pupils. The regression model consists of four steps that evolve from background variables to general and ICT-related teacher characteristics. In the first step, the variables '*Grade*' and '*Experience in education*' were entered (Model 1). Because Flemish early childhood education consists of three grades, a dummy variable was made with third grade as reference level. The grade and the number of years of experience in education accounted for 37.7% of the variance in the dependent variable. The grade and therefore the age of the children seems to be an important factor ($\beta = -.641, p < .001$). In the next step, the general teacher characteristics were added (Model 2). The addition of the variables '*Innovativeness*' and '*Teacher self-efficacy*' produced no significant added value in the proportion of variance explained ($\Delta R^2 = .002$). The general characteristics contribute little to the explanation of '*ICT use to support basic ICT skills and attitudes*'. Step 3 of the model consists of ICT-related teacher characteristics (Model 3). Both '*Teachers' ICT competences*' ($\beta = 0.164, p < .05$) and '*Experience with ICT at school*' ($\beta = .128, p < .05$) seem to have an important influence on the use of ICT in early childhood education, which leads to an increase in the R^2 with 7%. In the final model, the attitudes towards ICT were added (Model 4). These variables did not lead to a significant increase in explained variance in the dependent variable ($\Delta R^2 = .012$). In this case the third model is the most powerful, confirmed by the AIC and BIC, which reach their lowest value for Model 3. In this way, 44.9% of the variance in the dependent variable is explained.

ICT use supporting contents and individual learning needs

Table 6 presents the results of the hierarchical regression conducted to investigate which factors influence the second type of ICT use in early childhood education, '*ICT use supporting contents and individual learning needs*'. The model consists of the same four steps as the previous model. The first step immediately shows that the background variables lead to a lower proportion of variance explained ($R^2 = .127$), even though the variable '*Grade*' is still an important factor ($\beta = -.411, p < .001$) (Model 1). The addition of the general teacher characteristics (Model 2) did not lead to a significant contribution ($\Delta R^2 = .002$). The set of ICT-related teacher characteristics, however, contributed to a significant increase in the proportion of variance explained ($\Delta R^2 = .237$). Again, the '*Teachers' ICT competences*' are a significant factor ($\beta = .208, p < .05$). In this analysis, however, '*ICT professional development*' is also very important ($\beta = .355, p < .001$). This is a notable difference in

Table 4. Correlation matrix.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) Teacher self-efficacy										
(2) Innovativeness	.24**									
(3) Experience with ICT at home	.16*	.19**								
(4) Experience with ICT at school	.14*	.17**	.45**							
(5) Teachers' ICT competences	.23**	.23**	.30**	.13						
(6) ICT professional development	.18**	.10	.22**	.16*	.63**					
(7) Possibilities of ICT for teachers in early childhood education	.02	.03	.07	.08	.31**	.34**				
(8) Possibilities of ICT for preschoolers: ICT basic skills and attitudes	.12	.13	.13	.19**	.26**	.20**	.44**			
(9) Possibilities of ICT for preschoolers: development and learning process	.10	.07	.11	.08	.26**	.25**	.63**	.58**		
(10) ICT use in early childhood education: basic skills and attitudes	.07	.01	.04	.24**	.19**	.23**	.27**	.18**	.25**	
(11) ICT use in early childhood education: contents and individual learning needs	.08	-.01	.04	.15*	.38**	.46**	.48**	.25**	.33**	.58**

Note: * $p \leq .05$; ** $p \leq .001$.

Table 5. Hierarchical regression analysis with 'ICT use supporting basic ICT skills and attitudes' as dependent variable.

Standardised regression coefficients (β) ($n = 209$)				
Independent variables	Model 1	Model 2	Model 3	Model 4
<i>Background variables</i>				
Grade 1 (3 to 4 years)	-.641**	-.642**	-.630**	-.596**
Grade 2 (4 to 5 years)	-.085	-.085	-.102	-.084
Experience in education	.057	.065	.096	.104
<i>General teacher characteristics</i>				
Teacher self-efficacy		-.016	-.067	-.067
Innovativeness		.042	.016	-.008
<i>ICT-related teacher characteristics</i>				
Teachers' ICT competences			.164*	.144
ICT professional development			.115	.090
Experience with ICT at home			-.089	-.089
Experience with ICT at school			.128*	.123*
<i>Attitudes towards ICT</i>				
Possibilities of ICT to teach preschoolers ICT basic skills and attitudes				.054
Possibilities of ICT to support preschoolers in their development and learning process				.049
Possibilities of ICT for teachers in early childhood education				.043
R ² (proportion of variance explained)	.377	.379	.449	.461
Added value		.002	.070**	.012
AIC	1415.49	1418.95	1401.82	1403.13
BIC	1428.86	1439.00	1435.25	1446.58

Note: * $p \leq .05$; ** $p \leq .001$.

comparison to 'ICT use supporting basic ICT skills and attitudes'. Finally, the multiple scales with regard to the attitudes towards ICT were added (Model 4). The attitudes towards 'Possibilities of ICT for teachers in early childhood education' exert a significant influence on 'ICT use supporting contents and learning needs'. The final proportion of variance explained for this dependent variable amounts to 43.4%. The AIC and BIC show that for this analysis, the final model is the most powerful.

Conclusion and discussion

The present study aimed to develop a valid and reliable instrument to measure ICT use in early childhood education. With regard to this first research question, the factor analysis suggested a solution with two factors, indicating that two types of ICT use can be distinguished in preschools. This includes 'ICT use supporting basic ICT skills and attitudes' (N items = 5, $\alpha = .97$) and 'ICT use supporting contents and individual learning needs' (N items = 6, $\alpha = .86$). The first type of ICT use occurs more frequently than the second type ($M = 56.09$ and $M = 29.49$). These findings are consistent with an analysis of national and international technology curricula which indicate that two types of ICT use can be distinguished, ICT as a subject and ICT as an educational tool (Tondeur, van Braak, and Valcke 2007). When ICT is presented as a separate subject, the development of technological basic skills is the main aim. This is

Table 6. Hierarchical regression analysis with 'ICT use supporting contents and individual learning needs' as dependent variable.

Standardised regression coefficients (β) ($n = 209$)				
Independent variables	Model 1	Model 2	Model 3	Model 4
<i>Background variables</i>				
Grade 1 (3 to 4 years)	-.411**	-.314**	-.423**	-.351**
Grade 2 (4 to 5 years)	-.123	-.123	-.186*	-.130
Experience in education	-.046	-.053	-.003	.016
<i>General teacher characteristics</i>				
Teacher self-efficacy		.046	-.043	-.023
Innovativeness		-.021	-.044	-.049
<i>ICT-related teacher characteristics</i>				
Teachers' ICT competences			.208*	.173*
ICT professional development			.355**	.270**
Experience with ICT at home			-.105	-.100
Experience with ICT at school			.051	.056
Attitudes towards ICT				
Possibilities of ICT to teach preschoolers ICT basic skills and attitudes				.013
Possibilities of ICT to support preschoolers in their development and learning process				-.039
Possibilities of ICT for teachers in early childhood education				.295**
R ² (proportion of variance explained)	.127	.129	.366	.434
Added value		.002	.237**	.068**
AIC	1316.70	1320.20	1261.72	1243.99
BIC	1330.07	1340.25	1295.15	1287.44

Note: * $p \leq .05$; ** $p \leq .001$.

comparable with 'ICT use supporting basic ICT skills and attitudes'. The use of ICT as an educational tool is consistent with 'ICT use supporting contents and individual learning needs'. This type of use is based on an educational discourse and presumes ICT can support and improve learning or even fundamentally change education (Selwyn and Brown 2000). Although different authors (Plowman and Stephen 2003; Smeets 2005; Watson 2001; Yelland 2006) indicate that ICT can be a catalyst for educational change, they acknowledge the fact that ICT is rarely used in this way. ICT as an isolated activity to develop basic ICT skills generally seems to occur more than the use of ICT to support education and learning.

The second aim of this study was to explore which factors are related to ICT use in early childhood education. Firstly, the grade in which a preschool teacher works seems to be of great importance for both types of ICT use. In the second and third grade the use of ICT occurs more frequently and this is consistent with the concept of 'developmental appropriateness' (Bolstad 2004; Kalas 2010; Siraj-Blatchford and Siraj-Blatchford 2000). Within this concept, the teacher has the responsibility to judge which experiences with ICT are appropriate for a certain age and a certain child. In accordance with the teachers in the research of Wood et al. (2008), the participants in this study do not consider the introduction of ICT appropriate for children younger than three- to four-years-old.

Further, the general teacher characteristics are not related to the use of ICT in early childhood education. It seems that the participants in this study generally estimate their 'Inovativeness' and 'Teacher self-efficacy' highly, whether they use ICT in the classroom or not, as is the case in a study of Mueller et al. (2008). In the early nineties (Marcinkiewicz 1993) and later on (van Braak 2001; van Braak, Tondeur, and Valcke 2004), innovativeness proved to be an important variable in explaining the integration of ICT in the classroom. It is possible that ICT use is no longer perceived as an innovative practice. Therefore, it does not matter whether teachers are innovative or not. Other variables are currently more important in explaining why teachers use or do not use ICT within the context of early childhood education.

Within the ICT-related teacher characteristics, for example, important differences occur between 'ICT use supporting basic ICT skills and attitudes' and 'ICT use supporting contents and individual learning needs'. 'Teachers' ICT competences' exert an influence on both types of ICT use. The importance of teachers' self-perceived ICT competences can thus hardly be overestimated. Teachers need to feel they possess the necessary skills to use ICT in the classroom (Compeau, Higgins, and Huff 1999). However, 'Experience with ICT at school' is merely related to 'ICT use supporting basic skills and attitudes', whilst 'ICT professional development' is only related to 'ICT use supporting contents and individual learning needs'. As has been stated in the theoretical framework, ICT professional development is a crucial variable for ICT integration (Valcke et al. 2007; Vanderlinde and van Braak 2010). This study confirms that ICT professional development is a decisive factor in stimulating ICT use that transcends supporting basic ICT skills and attitudes.

With regard to the attitudes towards the possibilities of ICT in early childhood education, only attitudes towards 'Possibilities of ICT for teachers in early childhood education' are related to 'ICT use supporting contents and individual learning needs'. It is clear that the possibilities for preschoolers are estimated higher. It seems that, although teachers have positive attitudes towards the possibilities for preschoolers, other barriers inhibit the use of ICT. Teachers need to see the possibilities for themselves before they start integrating ICT in their classroom. Hew and Brush (2007) indicate that beliefs and attitudes are only one possible barrier. Other limitations include the available resources, knowledge and skills of teachers, and barriers at the school level, such as policy and leadership.

This study has certain limitations. Firstly, it solely questioned preschool teachers and did not include children's own experiences and perceptions of ICT (e.g. like Hatzigianni and Margetts 2012; McKenney and Voogt 2010; Morgan 2010). This study can be considered as a first step in exploring ICT use in early childhood education. Future research could give a more detailed account of teachers' practices and children's perceptions by using both quantitative and qualitative methods. Further, the variables used in this study are not exhaustive. Other general and ICT-related teacher characteristics could be included, such as general educational beliefs (e.g. Hermans et al. 2008), or a direct measurement of teachers' ICT competences. Moreover, this study solely focussed on teacher characteristics, since preschools are not obliged to use ICT. We therefore assumed that the current use of ICT could be attributed to efforts of individual teachers. Further research could include school-related variables, such as the availability of infrastructure and the presence of an ICT policy plan (e.g. Vanderlinde and van Braak 2010).

This study suggests that ICT professional development is a crucial factor in stimulating ICT use in early childhood education that transcends supporting basic ICT skills

and attitudes. Currently, Flemish preschool teachers are not obliged to use ICT in their classroom. Hence, some teachers use ICT to a great extent while others don't use it at all. Therefore, professional development initiatives should take into account the competence level and the actual use of ICT of preschool teachers. They can engage preschool teachers in conversations regarding why teachers use or do not use ICT in their classrooms. This way, non-users can be encouraged to discover the value of ICT in early childhood education. Further, they can address those teachers who already use ICT to support basic ICT skills and attitudes and stimulate them to lift their ICT use to a higher level. Also, teacher education can offer students an overview of different possibilities and technological tools. Pre-service teachers could be stimulated to try different ICT applications during teaching practice. In this way, they can build their own opinion regarding the use of ICT in their future practice. In our current society, where tablets in early childhood education are no longer unimaginable, and where the majority of preschoolers already have experiences with the Internet at home (Teuwens 2011), preschool teachers need to be challenged to think about the role that ICT can have in their classroom practice. Although the actual use of ICT in early childhood education is rather limited, the teachers who participated in this study do have positive attitudes towards the possibilities of ICT for preschoolers. This means that preschool teachers do not view ICT as a threat to playful learning and children's development, and that efforts should be made to support those who are willing to integrate ICT in their classroom practice.

References

- Bolstad, R. 2004. *The Role and Potential of ICT in Early Childhood Education: A Review of New Zealand and International Literature*. Wellington: Ministry of Education.
- van Braak, J. 2001. "Individual Characteristics Influencing Teachers' Class Use of Computers." *Educational Computing Research* 25: 141–157.
- van Braak, J., J. Tondeur, and M. Valcke. 2004. "Explaining Different Types of Computer Use Among Primary School Teachers." *European Journal of Psychology of Education* 19: 407–422.
- Campbell, A., and G. Scotellaro. 2009. "Learning with Technology for Pre-Service Early Childhood Teachers." *Australian Journal of Early Childhood* 34: 11–18.
- Clements, D. H., and J. Sarama. 2003. "Strip Mining for Gold: Research and Policy in Educational Technology – A Response to 'Fool's Gold.'" *AACE Journal* 11: 7–69.
- Compeau, D., C. A. Higgins, and S. Huff. 1999. "Social Cognitive Theory and Individual Reactions to Computing Technology: A Longitudinal Study." *MIS Quarterly* 23: 145–158.
- Cordes, C., and E. Miller. 2000. *Fool's Gold: A Critical Look at Computers in Childhood*. New York: Alliance for Childhood.
- Galanouli, D., C. Murphy, and J. Gardner. 2004. "Teachers' Perceptions of the Effectiveness of ICT-competence Training." *Computers & Education* 43: 63–79.
- Hatzigianni, M., and K. Margetts. 2012. "'I Am Very Good at Computers': Young Children's Computer Use and Their Computer Self-Esteem." *European Early Childhood Education Research Journal* 20: 3–20.
- Healy, J. M. 2003. "Cybertots: Technology and the Preschool Child." In *All Work and no Play: How Educational Reforms are Harming Our Preschoolers*, edited by S. Olfman, 83–111. Westport: Praeger Publishers.
- Hermans, R., J. Tondeur, J. van Braak, and M. Valcke. 2008. "The Impact of Primary School Teachers' Educational Beliefs on the Classroom Use of Computers." *Computers & Education* 51: 1499–1509.
- Hew, K. F., and T. Brush. 2007. "Integrating Technology into K-12 Teaching and Learning: Current Knowledge Gaps and Recommendations for Future Research." *Educational Technology Research and Development* 55: 223–252.

- Inan, F. A., and D. L. Lowther. 2010. "Factors Affecting Technology Integration in K-12 Classrooms: A Path Model." *Education Technology Research and Development* 58: 137–154.
- Kalas, I. 2010. *Recognizing the Potential of ICT in Early Childhood Education*. Moscow: UNESCO Institute for Information Technologies in Education.
- Marcinkiewicz, H. R. 1993. "Computers and Teachers: Factors Influencing Computer Use in the Classroom." *Journal of Research on Computing in Education* 26: 220–237.
- Marsh, J., G. Brooks, J. Hughes, L. Ritchie, S. Roberts, and K. Wright. 2005. *Digital Beginnings: Young Children's Use of Popular Culture, Media*. University of Sheffield: Literacy Research Centre.
- McKenney, S., and J. Voogt. 2010. "Technology and Young Children: How 4–7 Year Olds Perceive Their Own Use of Computers." *Computers in Human Behavior* 26: 656–664.
- Morgan, A. 2010. "Interactive Whiteboards, Interactivity and Play in the Classroom with Children aged Three to Seven Years." *European Early Childhood Education Research Journal* 18: 93–104.
- Mortelmans, D., and B. Dehertogh. 2008. *Factoranalyse*. Leuven: Acco.
- Mueller, J., E. Wood, T. Wiloughby, C. Ross, and J. Specht. 2008. "Identifying Discriminating Variables between Teachers Who Fully Integrate Computers and Teachers with Limited Integration." *Computers & Education* 51: 1523–1537.
- Plowman, L., and C. Stephen. 2003. "A 'Benign Addition'? Research on ICT and Pre-School Children." *Journal of Computer Assisted Learning* 19: 149–164.
- Plowman, L., and C. Stephen. 2005. "Children, Play, and Computers in Pre-School Education." *British Journal of Educational Technology* 36: 145–157.
- Plowman, L., and C. Stephen. 2006. *Technologies and Learning in Pre-School Education*. Paper presented at AERA annual meeting. Education Research in the Public Interest.
- Sang, G., M. Valcke, J. van Braak, and J. Tondeur. 2010. "Student Teachers' Thinking Processes and ICT Integration: Predictors of Prospective Teaching Behaviors with Educational Technology." *Computers & Education* 54: 103–112.
- Selwyn, N., and P. Brown. 2000. "Education, Nation States and the Globalization of Information Networks." *Journal of Education Policy* 15: 661–682.
- Siraj-Blatchford, I., and J. Siraj-Blatchford. 2000. *Children Using ICT: The Seven Principles for Good Practice*. Early Childhood – Developmentally Appropriate Technology. London. <http://www.datec.org.uk/guidance/DATEC7.pdf>
- Siraj-Blatchford, I., and J. Siraj-Blatchford. 2003. *More Than Computers: Information and Communication Technology in the Early Years*. London: The British Association for Early Childhood Education.
- Siraj-Blatchford, I., and J. Siraj-Blatchford. 2005. *ICT in the Foundation Stage: A Position Paper*. 327 Matters. <http://www.327matters.org/Docs/CfBT.pdf>
- Smeets, E. 2005. "Does ICT Contribute to Powerful Learning Environments in Primary Education?" *Computers & Education* 44: 343–355.
- Terreni, L. 2010. "Adding New Possibilities for Visual Art Education in Early Childhood Settings: The Potential of Interactive Whiteboards and ICT." *Australian Journal of Early Childhood* 35: 90–94.
- Teuwens, J. 2011. *Het gebruik van internet door preschoolers* [The use of the internet by pre-schoolers]. Unpublished Master's dissertation.
- Tondeur, J., J. van Braak, and M. Valcke. 2007. "Towards a Typology of Computer Use in Primary Education." *Journal of Computer Assisted Learning* 23: 197–206.
- Tondeur, J., M. Valcke, and J. van Braak. 2008. "A Multidimensional Approach to Determinants of Computer Use in Primary Education: Teacher and School Characteristics." *Journal of Computer Assisted Learning* 24: 494–506.
- Tschannen-Moran, M., and A. Woolfolk Hoy. 2001. "Teacher Efficacy: Capturing an Elusive Construct." *Teaching and Teacher Education* 17: 783–805.
- Tsitouridou, M., and K. Vryzas. 2004. "The Prospect of Integrating ICT into the Education of Young Children: The Views of Greek Early Childhood Teachers." *European Journal of Teacher Education* 27: 29–45.
- Valcke, M., I. Rots, M. Verbeke, and Johan van Braak. 2007. "ICT Teacher Training: Evaluation of the Curriculum and Training Approach in Flanders." *Teaching and Teacher Education* 23: 795–808.

- Vandenbroucke, F. 2007. *Competenties voor de Kennismaatschappij: Beleidsplan ICT in het onderwijs*. Brussel: Vlaams ministerie van onderwijs.
- Vanderlinde, R., and J. van Braak. 2010. "The E-Capacity of Primary Schools: Development of a Conceptual Model and Scale Construction from a School Improvement Perspective." *Computers & Education* 55: 541–553.
- Vanderlinde, R., J. van Braak, and R. Hermans. 2009. "Educational Technology on a Turning Point: Curriculum Implementation in Flanders and Challenges for Schools." *Education Technology Research and Development* 57: 573–584.
- Van Scoter, J., D. Ellis, and J. Railsback. 2001. *Technology in Early Childhood Education: Finding the Balance*. Latrobe: Northwest Regional Education Laboratory.
- Watson, D. M. 2001. "Pedagogy before Technology: Re-thinking the Relationship between ICT and Teaching." *Education and Information Technologies* 6: 251–266.
- Wood, E., J. Specht, T. Willoughby, and J. Mueller. 2008. "Integrating Computer Technology in Early Childhood Education Environments: Issues Raised by Early Childhood Educators." *The Alberia Journal of Educational Research* 54: 210–226.
- Yelland, N. 2005. "The Future is Now: A Review of the Literature on the Use of Computers in Early Childhood Education (1994–2004)." *AACE Journal* 13: 201–232.
- Yelland, N. 2006. "Changing Worlds and New Curricula in the Knowledge Era." *Educational Media International* 43: 121–131.

Appendix 1: Construction of the independent variables ‘Attitudes towards the possibilities of ICT in early childhood education’

The results of the factor analysis on attitudes towards possibilities of ICT for preschoolers indicate a solution with two factors (Table A1). The first factor measures the attitudes towards ‘Possibilities of ICT to support preschoolers in their development and learning process’. The second factor indicates the attitudes towards ‘Possibilities of ICT for preschoolers to learn ICT basic skills and attitudes’. This distinction fits the identified types of ICT use in early childhood education.

Table A1. Factor analysis for attitudes towards ‘Possibilities of ICT for preschoolers’.

	Factor	
	1	2
The use of ICT in early childhood education promotes active participation of preschoolers in the learning process.	.80	
The use of ICT in early childhood education offers possibilities to support the general development of preschoolers.	.73	
The use of ICT in early childhood education increases the motivation of preschoolers.	.73	
The use of ICT in early childhood education offers possibilities to support the learning of preschoolers.	.66	
Preschoolers with learning difficulties can get many advantages out of the didactical possibilities of ICT.	.60	
The use of ICT in early childhood education increases the creativity of preschoolers.	.53	
Preschoolers can develop a positive attitude towards ICT when ICT is used in early childhood education.		.95
Preschoolers can learn basic ICT skills (e.g. mouse click) when ICT is used in early childhood education.		.65
Preschoolers learn about new media when ICT is used in early childhood education.		.56
R^2 (proportion of variance explained)	33.4%	23.3%
α	.80	.86

For attitudes towards ‘Possibilities of ICT for teachers in early childhood education’ one factor was found (Table A2). This scale includes tasks which can be executed more easily due to ICT.

Table A2. Factor analysis for attitudes towards ‘Possibilities of ICT for teachers in early childhood education’.

	Factor 1
Thanks to ICT I can better support the learning process of preschoolers.	.88
Thanks to ICT I can better remediate in class.	.87
Thanks to ICT I can better support the development process of preschoolers.	.85
Thanks to ICT I can better differentiate in class.	.84
Thanks to ICT I can better group preschoolers.	.76
Thanks to ICT I can give more directed feedback to preschoolers.	.75
Thanks to ICT I can better explain or illustrate certain things to preschoolers.	.71
Thanks to ICT I can put children to work in a more independent way.	.66
Thanks to ICT I can teach information skills to preschoolers.	.62
R^2 (proportion of variance explained)	60.0%
α	.93